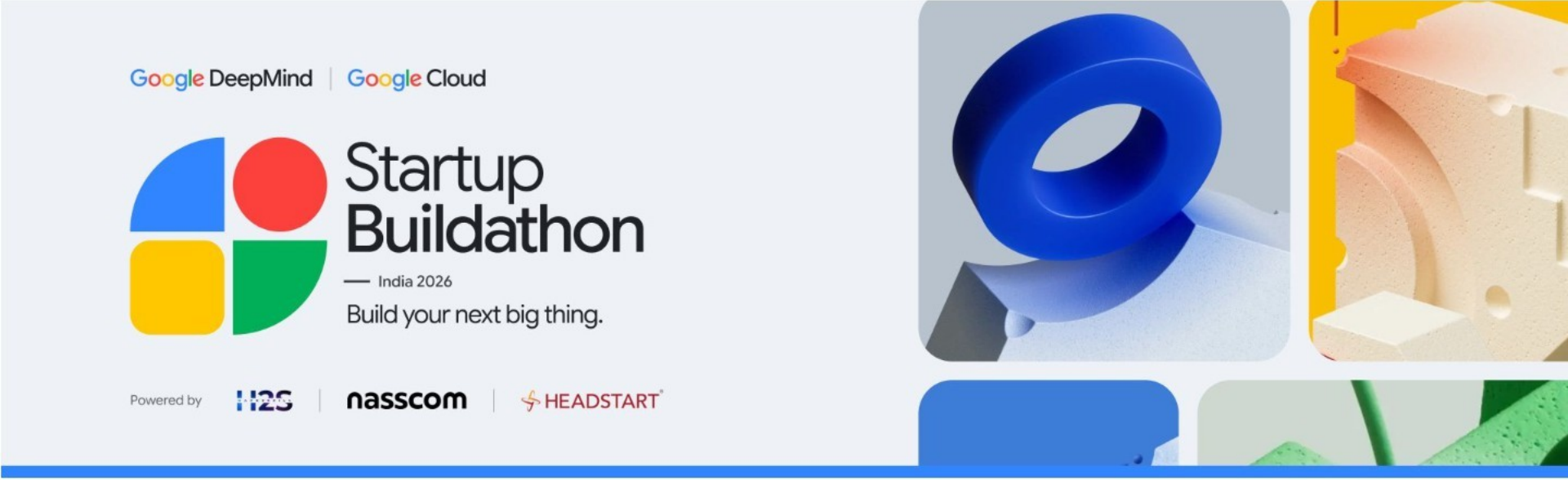


Team Name : StackScore

Team Size : Solo

Problem Statement : Hire for how engineers actually work using an AI-powered Full-Stack assessment framework

stackscore

Live App | Demo Video

Problem Statement

Current technical hiring platforms measure what candidates memorize - not how they engineer to solve a real-time problem.

01

Wrong Signal

LeetCode/Hackerrank style tests measure algorithmic recall and practice-driven pattern matching. They say nothing about system design judgment, requirements clarity, or adaptability under pressure - the skills that define a strong full-stack engineer.

A candidate who aces dynamic programming may not design a URL shortener that survives a 10× traffic spike.

02

No Standard Rubric

Every panel interviews differently. Results are undocumented, uncomparable across roles or companies, and irreproducible. The same candidate passes or fails based on who happened to be available - not on structured capability.

Hire decisions reduce to interviewer intuition, not documented engineering capability.

03

High Cost, Low output

A live system-design interview costs 2–3 senior engineering hours per candidate. No feedback is given to candidates. HR can't track or compare scores over time. There's no data layer for improving the process.

Companies spend hours per candidate and get a binary yes/no with no reusable signal.

Proposed Solution

stackscore

STAGE 1 . 5min
Clarify
Candidate Q&A with AI stakeholder to scope a deliberately ambiguous brief

→

STAGE 2 . 10min
Propose
Written system design against requirements uncovered in Stage 1

→

STAGE 3 . 20min
Build
Code the core module; runs in a sandboxed Node.js executor

→

STAGE 4 . 10min
Adapt
AI injects a mid-task constraint twist; candidate updates design & code

→

STAGE 5 . 15min
Debug
Candidate finds and fixes a subtle defect AI seeded into their own code

25 scenarios from a 5 × 5 matrix: problem domains × engineering constraint pillars

Gemini 3.8 Flash generates · **Claude Opus 5.5** judges (adaptive thinking)

Output: **StackScore 0–100** · role-calibrated · shareable URL

Scoring Rubric

Solution Design / Approach 55 %

D1 Requirement Framing
Quality of clarifying questions; did they scope the real ambiguity in Stage 1?

D2 Architecture & Component Design
Soundness and clarity of the proposed structure

D3 Tradeoff Reasoning
Explicit justification of choices against the assigned constraint

D4 Constraint Satisfaction
Does the design actually hold up against the assigned pressure - not just mention it?

D5 Adaptability
How well the design absorbs the Stage 4 twist: patched vs re-architected vs collapsed

Each axis independently scored 0–100 · **StackScore** = Σ(axis × weight)

Code-Level Best Practices 45%

K1 Design-Code Alignment
Does the implementation match Stage 2 proposal? Flags drift.

K2 Correctness & Edge Cases
Does it work, including boundary and failure conditions?

K3 Code Quality & Maintainability
Naming, modularity, separation of concerns

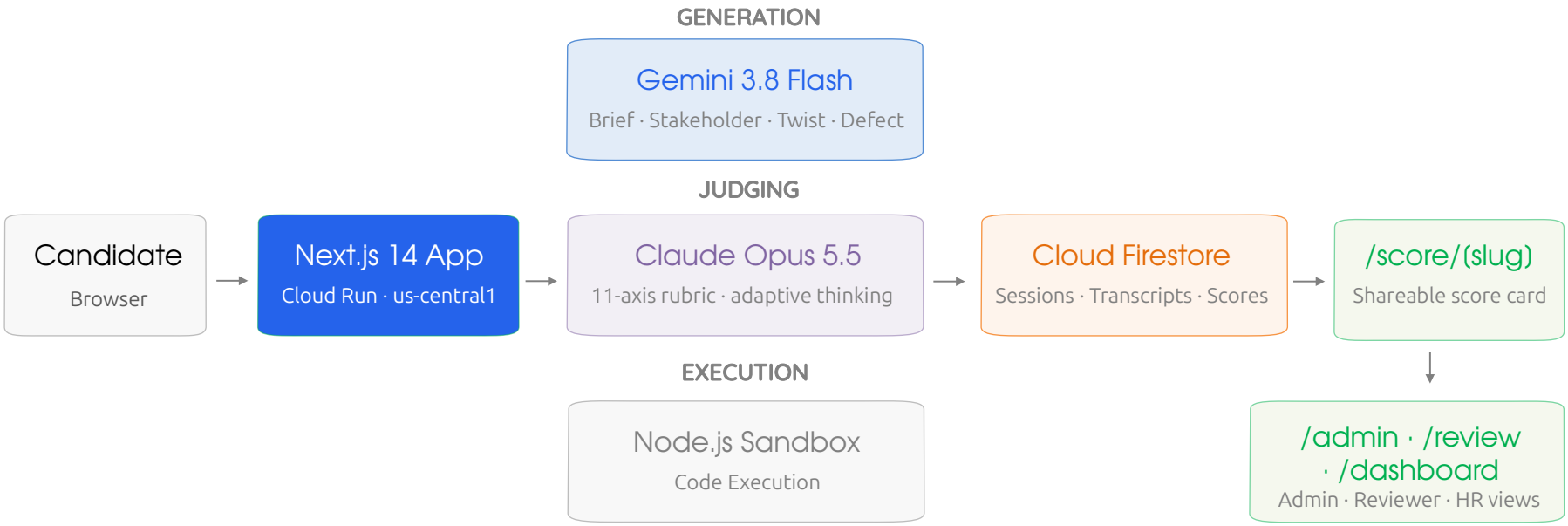
K4 Security & Defensive Coding
Input validation, secrets handling, auth checks

K5 Testing & Observability
Test coverage, logging, debuggability

K6 AI Collaboration Quality
how effectively candidates leverage AI tools - quality of prompting, verification, and knowing when to override.

Role-calibrated: **Junior** · **Mid** · **Senior** · Axis-independent scoring prevents halo bias

Solution Architecture



Google AI Tools

01
Google AI Studio

Gemini API key provisioned via AI Studio. Used in production for all generation calls - no Vertex AI billing complexity needed for the MVP.

aistudio.google.com

02
@google/genai SDK

Official JS SDK for Gemini API. genai.models.generateContent() powers brief generation, stakeholder voice, twist injection, and defect seeding.

model: "gemini-3.8-flash"

03
Cloud Run

Serverless container hosting for the Next.js app. Scales to zero between sessions; 512 MiB RAM, deployed automatically via Cloud Build CI/CD.

us-central1 · auto-scaled · zero-ops

04
Cloud Firestore

Document database storing sessions, stage-by-stage transcripts, and score cards. Queried in the HR dashboard for pass/fail threshold filtering.

sessions · transcripts · scores

05
Cloud Build

CI/CD pipeline via cloudbuild.yaml. Builds Docker image, pushes with COMMIT_SHA tag, and deploys to Cloud Run in one gcloud builds submit command.

cloudbuild.yaml · Docker · Cloud Run

06
Secret Manager

Stores GEMINI_API_KEY, ANTHROPIC_API_KEY, FIREBASE_PRIVATE_KEY. Injected into Cloud Run at deploy via --set-secrets. Zero secrets in source code.

--set-secrets KEY=secret:latest

Google AI Models

Gemini 3.8 Flash — GENERATION

via @google/genai · API Key mode

- ✓ Problem brief generation - deliberately underspecified, stakeholder-framed, with hidden constraint woven in implicitly
- ✓ Stakeholder voice simulation - answers clarifying questions in character without revealing the constraint category
- ✓ Mid-task constraint twist injection - a natural "scope change from leadership" reframing the session's pressure
- ✓ Debug defect seeding - plants one non-trivial logic error (off-by-one, missing edge case) in the candidate's own code

```
genai.models.generateContent({ model: "gemini-3.8-flash", contents: prompt })
```

Claude Opus 5.5 — JUDGING

via Anthropic SDK · adaptive thinking

- ✓ Rubric-constrained 11-axis evaluation - each axis scored 0–100 independently to prevent holistic-impression bleed
- ✓ thinking: { type: "adaptive" } lets the model reason through nuanced design tradeoffs before committing to scores
- ✓ Role-calibrated prompt - Junior/Mid/Senior expectation bars stated explicitly so the same rubric spans all levels
- ✓ Returns strict JSON: axis scores + 1–2 sentence feedback per axis, persisted to Firestore and displayed on the score card

```
thinking: { type: "adaptive" }, model: "claude-opus-5-5", max_tokens: 16000
```

Fallback Judge — RESILIENCE

via Anthropic SDK · Judge fallback

- ✓ Tier 1: Claude Opus 5.5, calibrated — rubric + anchors + adaptive thinking, full fingerprint check
- ✓ Tier 2: Claude Opus 5.5, uncalibrated — triggered on fingerprint mismatch; session gets "Partially Judged" badge and is enqueued for rejudge drain
- ✓ Tier 3: uncalibrated also fails → No model — scoring hold active or model unavailable; session goes to pending_review and a human reviewer task is created. human reviewer scores blind (rubric shown, LLM scores hidden to prevent bias)

Feature Highlights

What was built for this buildathon beyond the core assessment towards MVP.

01 Judge Fallback Pipeline

Three-stage resilience: calibrated judge → uncalibrated retry ('Partially Judged' badge) → auto-created human review task. No assessment is ever silently lost. Rejudge queue with retry limits and automatic escalation to human review when exhausted.

calibrated → partial → pending_review

02 Human Reviewer Role

Fourth auth role with its own cookie and registry. Reviewer sees the rubric with anchor bands but NOT the LLM scores — bias-free by design. Admin assigns tasks; reviewer submits structured scores that replace axis scores on the score card.

ss_reviewer cookie · bias boundary at /api/review/submission

03 Scoring Hold & Queue

Admin can pause the scoring pipeline with a toggle — flag in Firestore, not an env var. Submissions during a hold are queued in Firestore and drained automatically when unpaused. Alert-only by default; hold is opt-in per admin decision.

Firestore-backed · admin toggle · drain on resume

04 Calibration Monitor

Daily Cloud Scheduler probe: runs 3 hard-assertion fixtures (~$0.40), checks calibration fingerprint (rubric hash + model + effort), alerts if stale. Prevents silent calibration drift after a rubric edit or model change without re-running the full sweep.

daily · fingerprint: sha256(rubric + model + effort)

05 Gemini Cross-Judge

Scripts to send contested fixtures to Gemini 3.8 Flash and compare scores. On K2 (correctness): Gemini confirmed our judge's scoring on 4/5 contested cells. On K5 (testing): both judges agreed near the band floor — authored bands were optimistic.

gemini-triage.json · K2 confirmed · K5 bands revised

06 Full Instrumentation

Every score record stores model, effort, latency, token counts (input/output/cache), and estimated cost per assessment (~$0.14 at Opus 5.5 pricing). Data layer ready for 10x/100x scaling analysis and cost optimization.

judgeUsage · ~$0.14/assessment · cache-aware

Generation Resilience & Scenario Bank

Four-tier fallback ensures zero candidate-facing downtime during Gemini outages.

01 4-Tier Generation Chain

Gemini 3.8 Flash → Claude Haiku 5.5 → Scenario Bank → Static template. Each tier falls back automatically, transparent to the candidate. Live generation is always tried first; the bank is a last-resort backstop that keeps the interview running.

Gemini Flash → Haiku 5.5 → bank → template · automatic

02 Pre-Generated Scenario Bank

20 fully-formed assessment packs (brief + twist) pre-generated and stored in Firestore. Cloud Scheduler refills the pool every 2 hours. Packs span all 25 scenario matrix cells (5 domains × 5 constraint pillars).

scenarioBank collection · target pool: 20 · 25-cell coverage

03 Atomic Bank Allocation

Firestore runTransaction ensures two concurrent sessions can never claim the same pack. Status flips available → allocated atomically. Race-free under any concurrency level; no double-booking possible.

runTransaction · available → allocated · zero races

04 __pending__ Orphan Reclaim

When session creation fails mid-flow, the allocated pack is marked __pending__. Cloud Scheduler runs reclaim-bank-packs every 30 min to return orphaned packs (10-min TTL) to the available pool. No pack is permanently lost.

__pending__ sentinel · 10-min TTL · 30-min reclaim cycle

05 Twist Pre-Generation

Bank packs include a pre-generated twist alongside the brief. inject-twist pulls the stored twist directly — no AI call at Stage 4. Twist injection stays instant even during a complete Gemini + Claude outage.

bankTwist stored on session · zero-latency Stage 4 · outage-safe

06 Cloud Scheduler Automation

fill-scenario-bank (every 2h) tops up the pool. reclaim-bank-packs (every 30 min) returns orphaned packs. Admin dashboard at /admin shows live bank stats (available / allocated / total) with one-click manual refill.

fill every 2h · reclaim every 30min · /admin dashboard

Reliability Evidence

What the judge can prove — without human graders, without authored bands.

01

Rank Stability

Three independent runs produced the identical ordering of all 10 submissions. Spearman ρ = 1.000 with zero pairwise inversions across all 45 "is A stronger than B?" comparisons. Ranking is the deliverable — this is the number that matters commercially.

ρ = 1.000 · 0 inversions / 45 pairs · 3 runs

02

Separation

Every adjacent pair of submissions is separated by ≥ 2.3 standard deviations (pooled run-to-run sd = 1.18 pts). Gaps range 2.3–14.4 sd. Reliability without separation is useless — a judge scoring everyone 50 is perfectly consistent and completely uninformative.

2.3–14.4 sd gap · pooled sd 1.18 pts · 10 submissions

03

Rule Compliance

5/5 hard-assertion fixtures passed: rule-based requirements stated in the system prompt are obeyed including a prompt-injection attempt (scored 12/100 with K4=2). Gemini 3.8 Flash independently scored in-band on 4/5 contested K2 cells, confirming our judge's correctness axis is directionally sound.

5/5 assertions · Gemini cross-validated · panel protocol in repo

Market Positioning & USP

Why StackScore is different from every tool already on the market.

01
vs LeetCode / HackerRank

Measures algorithmic recall and speed. Passes engineers who can't design systems; fails engineers who can. Zero signal on requirements framing, tradeoff reasoning, adaptability under pressure, or AI collaboration quality. StackScore: 5-stage scenario, 11 axes, role-calibrated, no memorization advantage.

ρ = 1.000 · 0/45 inversions · ≥ 2.3 sd gap

02
vs CodeSignal / Codility

Pre-employment coding tests with some system-design add-ons. AI judging is "AI-powered" in name — no published calibration, no reliability evidence, no audit trail. Scores are black boxes with no rubric transparency. StackScore: open rubric with anchor bands, published sweep results, human fallback.

5 stages · 25 scenarios · 11 axes · role-calibrated

03
vs Karat / Human Panels

Gold standard for signal quality but costs 3–5 engineering hours per candidate. Doesn't scale. No structured rubric → results are undocumented and non-comparable. StackScore: same holistic judgment, structured into 11 axes, delivered in 60 min at ~$0.14 per assessment with a shareable, verifiable score URL.

open rubric · sweep in repo · panel protocol committed

04
Evidence-Based Reliability

The only publicly demonstrated reliability measures in this space: Spearman ρ = 1.000 (zero rank inversions across 3 independent runs), every adjacent pair ≥ 2.3 sd apart. Cross-validated by a second major model (Gemini). Accuracy claim deliberately withheld — protocol in repo, not in a press release.

vs ρ = unknown · black-box AI · no audit trail (competitors)

05
Scenario Depth + Adaptability

5-stage scenario (Clarify → Propose → Build → Adapt → Debug) captures the full engineering workflow, not a single snapshot. The mid-task twist and seeded debug defect are unique stress tests not present in any competitor. The 25-scenario matrix (5 domains × 5 constraint pillars) means every session is a different problem.

5 domains × 5 pillars · twist + debug stages unique

06
Calibration Infrastructure

Rubric anchors, calibration sweeps, panel protocol, monitor scheduled daily — all committed to the repo. Any user of the platform can inspect the rubric, run the sweep, and validate the judge. No competitor publishes their rubric, their calibration, or their reliability measures. That is structural differentiation.

data/calibration/ · PANEL.md · daily monitor live

Google DeepMind | Google Cloud

Startup Buildathon

— India 2026

Build Your Next Big Thing.

Powered by H2S | nasscom | HEADSTART

Thank you





